CONNECTDESK · PHASE 2 · RECOMMENDATION FOR DECISION

Calling without the PBX: what Fast Dial should do about Voiso, and how we build Phase 2

Fast Dial asked whether to finalise its contract with Voiso or wait for something better. Our advice: **do not sign an annual contract yet.** Test Voiso for two weeks on Fast Dial's own numbers while ConnectDesk builds **direct calling from the browser**, with no Yeastar, MicroSIP or 3CX, and working from home too. Keep the PBX as the default throughout, and decide on **Friday 6 November** with measured results.

For	From	Date	Decision date proposed
The CEO, Fast Dial	ConnectDesk	Thursday 8 October 2026	Friday 6 November 2026

Part A (sections 1–6) is for the CEO. Part B (sections 7–12) is for the people who build and run it. Bring Appendix A to the Voiso demo. Sources are in Appendix D. Prices are as published on 8 October 2026, at KES 129 to the US dollar.

PART A · FOR THE CEO

1 The answer on one page

OUR RECOMMENDATION

1. **Hold the Voiso signature, but keep talking to Voiso.** Voiso is a real contact-centre product. Its own documentation shows a browser softphone, a predictive dialler, and supervisor *Monitor*, *Whisper*, *Barge-in* and *Intercept*. So yes, it can call from the softphone with every in-call supervisor action. It has not yet shown that it works *in Kenya, at Kenyan prices, with Fast Dial's caller ID and with lenders' data*. Its monthly plans need no contract, so test it for two weeks against Appendix A before committing to a year.
2. **At the same time, build direct calling into ConnectDesk.** The Riri dock in the agent's browser calls the customer through our own calling engine (a *voice node*) in Nairobi and a Kenyan SIP trunk. Agents can work from the office or from home. Most of the parts already run in Phase 1: the browser phone, listen/whisper/barge, recording and Riri's inbound line.
3. **Keep the PBX as the default.** Each organisation, campaign and agent gets a *phone route* setting: PBX (default), Direct, or Voiso if Fast Dial signs. Switching back is one click, so nothing in Phase 2 can stop the floor.
4. **Run ConnectDesk from Nairobi, next to its data.** At 30 agents, hosting on Vercel would cost about $300–700 a month (KES 39,000–90,000). Two Linux servers in the same Nairobi data centre as the CallBox server cost about KES 16,000–20,000 a month, and they make every screen faster.
5. **Decide on Friday 6 November.** Put all three routes through one scorecard (section 4) using measured calls, not demos.

The five things Voiso has to prove before Fast Dial signs

1. **A legal Kenyan caller ID.** Kenyan networks block calls that arrive from abroad showing a +254 number. In 2022 the Communications and Multimedia Appeals Tribunal upheld the regulator's ban on

disguising international calls as local ones. Unless Voiso starts Fast Dial's calls *inside Kenya*, customers see a foreign number or the call never arrives. Voiso would have to use a Kenyan carrier or Fast Dial's own trunk.

2. **Kenyan minute prices.** International cloud routes charge $0.39–0.45 a minute to Safaricom and Airtel numbers (Twilio and Plivo publish these rates). A Kenyan trunk costs roughly KES 2.50–4. At Fast Dial's volume of about 30,000 minutes a month, that is about **KES 1.5–1.7 million** against **KES 75,000–120,000**.
3. **Where the audio travels.** Voiso lists six data centres, in North America, Europe and Asia, and none in Africa. When a Nairobi agent calls a Nairobi customer, both voices may cross to another continent and back. That adds delay that callers notice.
4. **One screen for the agent.** Voiso's API starts and stops calls and returns call records. Its documentation shows no API for hold, transfer or listen/whisper/barge, so supervisors would supervise in Voiso, not in ConnectDesk's command centre. Agents would work across two screens.
5. **Lenders' data.** Borrowers' call recordings would sit outside Kenya unless Voiso copies them to storage Fast Dial controls (its Pro plan offers copies to Amazon S3). Each lender's data agreement must allow that.

2 Why the PBX holds the floor back

Phase 1 works. Fast Dial confirmed today that click-to-call, the auto-dialler and supervisors' listen, whisper and barge all work in ConnectDesk. They work *through the Yeastar S50*, though, and the S50 is the weak link:

WHAT WE MEASURED IN PHASE 1	WHAT IT MEANS FOR THE FLOOR
On Monday 5 October the S50's control port went dark for **5 hours 21 minutes** (10:33–15:54). Agents kept dialling by hand, but ConnectDesk could not place calls.	One box in the office can stop the whole floor.
The S50 restarted about **1.5 times per working day** even before ConnectDesk went live, going dark for 2–8 minutes each time.	Calls in progress at a restart are dropped and leave no record.
It carries at most **25 calls at once**.	A predictive dialler for 20 agents needs more lines than that.
Only its control port is reachable from the internet. Each agent needs MicroSIP on a PC on the office Wi-Fi (the 3CX app cannot register to a Yeastar).	No agent can work from home without network changes.
Inbound calls went to an extension with no phone: **4,522 calls in 14 days, 2 answered** (late September). Riri now answers them, but only while a laptop on the office Wi-Fi runs her.	Answering customers depends on one laptop.

Phase 2 builds on what Phase 1 already runs. ConnectDesk talks to any phone system through one interface, so a Direct engine or a Voiso adapter plugs in behind the same screens. A browser phone already registers to an Asterisk *media edge*, where listen/whisper/barge, recording and Riri's keypad menu already run. The floor, the dialler and its contact policy, recordings, quality reviews and client reports stay as they are.

3 Voiso, examined

Voiso was founded in Singapore in 2018 and sells a cloud contact centre. It has over 300 staff and six data centres, and it is strongest in fintech and forex sales. We read its documentation and pricing on 8 October 2026.

CAPABILITY	WHAT VOISO'S OWN DOCUMENTATION SAYS	OUR READING
Browser softphone	The *Agent Panel*, a WebRTC softphone. Google Chrome is the only supported browser.	Meets the "call from the softphone" need.
Supervisor tools	Monitor, Whisper, Barge-in and Intercept, from the *Agent Zoom* view. The call opens in the supervisor's own Agent Panel. Agents on desk phones can be supervised with dial codes.	Complete. Supervision happens inside Voiso.
Dialler	AI predictive and progressive dialler, with answering-machine detection (AMD). Pro plan.	Mature. Pacing must be tuned for Kenyan networks.
Your own carrier	"Voice providers" (bring your own carrier) over a SIP trunk, authenticated by username/password or by static IP, over UDP, TCP or TLS, with optional SRTP.	The only route to a legal Kenyan caller ID at Kenyan prices.
API & webhooks	Click-to-call *start* and *stop*; call detail records; add and update contacts in dialler campaigns; users, teams and queues; webhooks retried on failure. Pro plan.	Enough for screen pops and outcomes. No documented API for hold, transfer, mute or monitoring.
Recordings	Kept 6 months (Core) or 1 year (Pro). Can be copied to the customer's Amazon S3 (Pro).	Recordings could be kept under Fast Dial's control.
Embedding	The agent workspace can be embedded in another web app as an iframe.	Could sit inside ConnectDesk. Needs testing.
Price	Core $30 per user per month ($24 billed annually). Pro $65 ($52 annually). "Unlimited telephony" $149 per user per month. Calls, AMD, caller IDs and numbers are billed on top. 14-day trial. No contract on monthly plans.	Fast Dial needs Pro, about KES 184,000 a month for 22 users before minutes.

What its documentation does not show: any African data centre; Kenya on its local caller ID page ("local numbers in over 120 countries"); a per-minute rate for Kenya ("destination-specific"); an API for hold, transfer, mute or supervisor monitoring; or any noise cancellation. Competitors such as Vonage and NICE CXone sell separate noise removal for the agent's side and the customer's side, plus removal of background voices.

> VERDICT
>
> Voiso can do what Fast Dial asked: a browser softphone with listen, whisper and barge. So the question is not *whether it can*, but **whether it can in Kenya, at a sensible cost per minute, without splitting the agent's work across two screens**. A two-week test answers that. Appendix A has the questions and the tests.

4 The options side by side

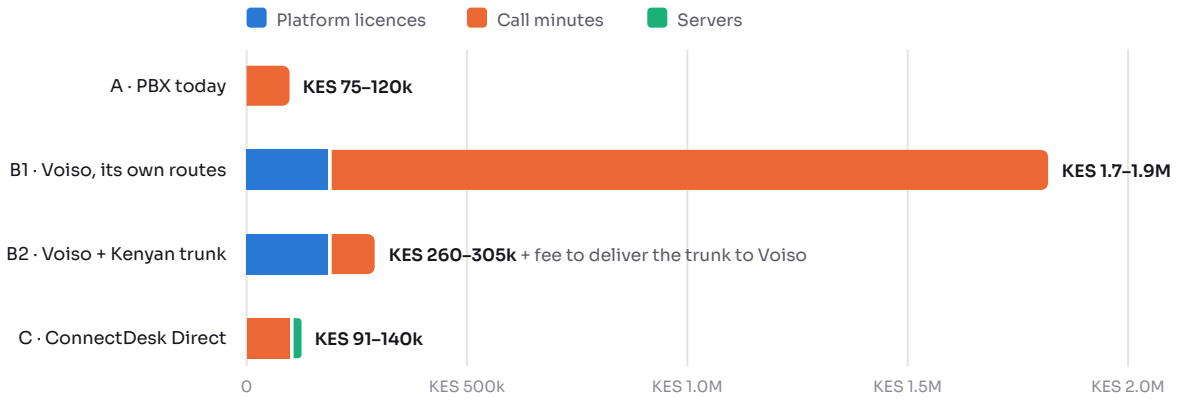
OPTION	WHAT IS GOOD	WHAT STOPS IT	ROLE IN THE PLAN
A · Stay on the PBX	Works today. Fast Dial owns it, with its own trunk and numbers.	Outages and restarts. 25-call ceiling. No remote agents. MicroSIP on every PC.	Default and fallback
A2 · Newer PBX Yeastar P-Series	Yeastar's browser client, call-centre panel and auto-dialler come only with the P-Series. Remote access is built in.	Still a PBX with its own licence and API. ConnectDesk needs a new driver. Does not fix noise.	Plan B if Direct fails
B · Voiso	A complete product with a mature dialler and every supervisor tool.	The five proofs in section 1: caller ID, minute price, audio path, one screen, data.	Two-week test; adapter if chosen

OPTION	WHAT IS GOOD	WHAT STOPS IT	ROLE IN THE PLAN
C · ConnectDesk Direct	Kenyan trunk, so a legal caller ID at local prices. Servers in Nairobi. One screen. Reuses Phase 1. Open-source engine with no per-seat fee. Data stays in Kenya.	We run it, so the risk is ours to build and operate. Carriers take time to deliver a trunk.	**Build now**; the PBX fallback contains the risk
D · Voice API Africa's Talking, Twilio	Africa's Talking is Kenyan, with Kenyan numbers, KES prices and a browser SDK.	No whisper or barge in these APIs. Twilio charges $0.39 a minute to Kenyan mobiles.	Africa's Talking as a *carrier*, not the engine
E · Build from scratch FreeSWITCH, jambonz, LiveKit, VICIdial	Full control.	Months of telephony plumbing before agents see any value (see section 9). jambonz now needs a paid licence to self-host. LiveKit cannot register to a carrier, and its phone-call noise removal runs only in its cloud. VICIdial would replace ConnectDesk instead of serving it.	No

The scorecard for 6 November

MEASURE	TARGET	HOW WE MEASURE IT
Caller ID on the customer's phone	**Must pass:** Fast Dial's own Kenyan number, on Safaricom and Airtel handsets	Test handsets, 20 calls each
Cost per connected minute	Lowest wins: licences + minutes + servers, divided by connected minutes	Invoices and call records
Connected calls per agent hour	Higher is better, on the same campaign	ConnectDesk reports
Audio	Average MOS 4.0 or better; one-way delay under 150 ms (ITU-T G.114); no one-way audio	Per-call quality statistics
Supervisor tools in ConnectDesk	Listen, whisper and barge from the command centre, connected within 3 seconds	Live test
Availability	99.5% or better during working hours	External monitoring
Recordings and data	Kept in Kenya, or abroad only with each lender's written consent	Contracts
One screen for agents	Yes or no	Watching the floor

5 What each option costs a month



Monthly running cost for 20 agents and 2 supervisors. Bars show the middle of each range; the table below gives the ranges. Assumes about 30,000 billable minutes a month to Kenyan mobiles. In September Fast Dial answered about 1,000 outbound calls a day, and the floor talked for about 94 hours in the week to 6 October. ConnectDesk's own subscription is the same in every option and is left out. Workings are in Appendix C.

OPTION	PLATFORM LICENCES	CALL MINUTES	SERVERS	TOTAL A MONTH
A · PBX today	None (the S50 is owned)	KES 75–120k at Fast Dial's trunk rate (to confirm)	None	KES 75–120k
B1 · Voiso, its own routes	22 × $65 = KES 184k	30,000 × $0.39–0.45 = KES 1.51–1.74M, and a foreign or blocked caller ID	None	KES 1.7–1.9M
B2 · Voiso + Kenyan trunk	KES 184k (KES 148k billed annually)	KES 75–120k, plus the carrier's fee to deliver the trunk to Voiso	None	KES 260–305k + fee
C · ConnectDesk Direct	None (open-source engine)	KES 75–120k	KES 16–20k (+ optional AI noise removal, about KES 26k)	KES 91–140k

One-off cost in every option: noise-cancelling USB headsets, about KES 6,000–15,000 each (section 8). B1's minute rate is the published rate of Twilio and Plivo to Kenyan mobiles. Voiso does not publish its Kenyan rate, so getting it is the first question in Appendix A.

6 What we need from Fast Dial this week

- [ ] **Voiso:** tell them monthly terms only for now. Send Appendix A and book a technical demo with their *telephony engineer*, not only sales. Ask for a two-week trial on Fast Dial's own numbers.
- [ ] **Carriers:** ask Safaricom, and one more carrier (Airtel or Africa's Talking), to quote a SIP trunk: 30 channels, delivered to a Nairobi data centre on a static IP, with Fast Dial's numbers as caller ID. Ask for the rate per minute by network, the billing increment, the set-up and monthly fees, and the lead time.
- [ ] **Baseline:** share the current Safaricom trunk bill (rate, channels and minutes). Every comparison starts from it.
- [ ] **People:** name 5 pilot agents (3 in the office and 2 at home) and 1 supervisor, and approve 5 noise-cancelling headsets.
- [ ] **Diary:** hold Friday 6 November, 10:00, for the decision.

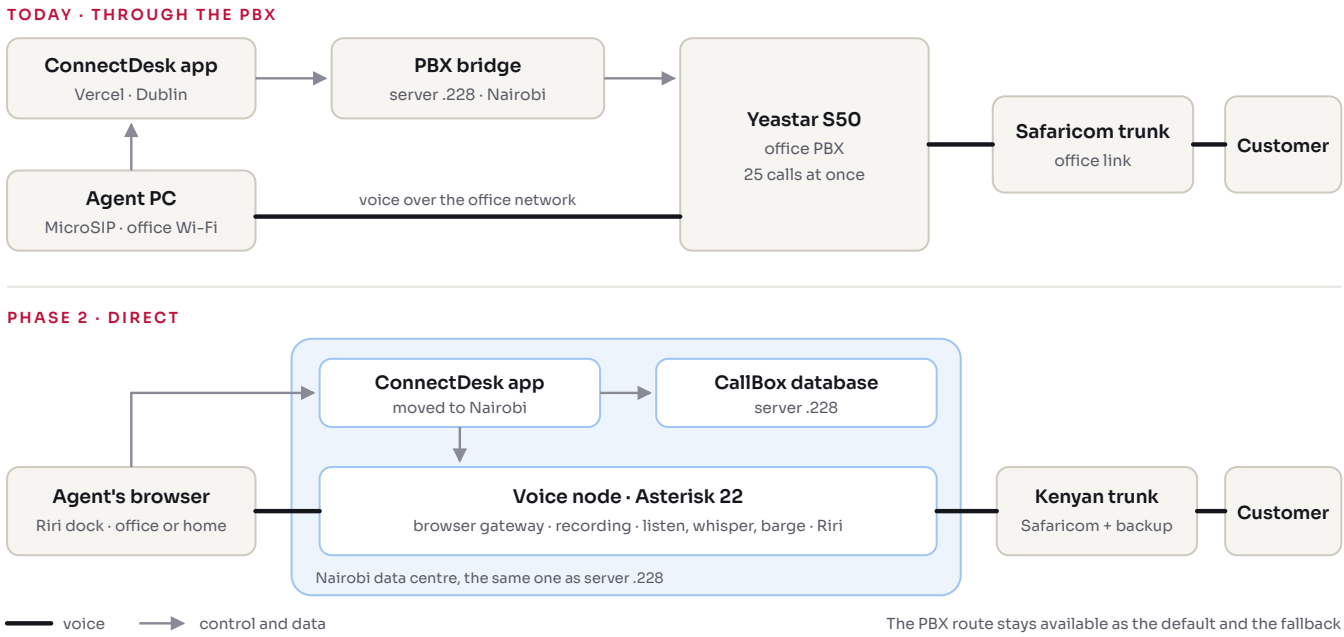
> WORDS USED IN THIS DOCUMENT
>
> **PBX**: the office phone exchange (Fast Dial's Yeastar S50). **SIP trunk**: a carrier's digital phone line, delivered over a network instead of copper. **Caller ID (CLI)**: the number the customer sees. **BYOC**: "bring your own carrier", plugging your own trunk into someone else's platform. **WebRTC**: the standard that lets a browser make calls. **TURN**: a relay that gets audio through strict home routers. **AMD**: answering-machine detection. **MOS**: a 1–5 call-quality score; 4.0 and above sounds clear. **Voice node**: our calling engine (Asterisk) in Nairobi.

PART B · FOR THE BUILD TEAM

How we build it, run it, and switch it on

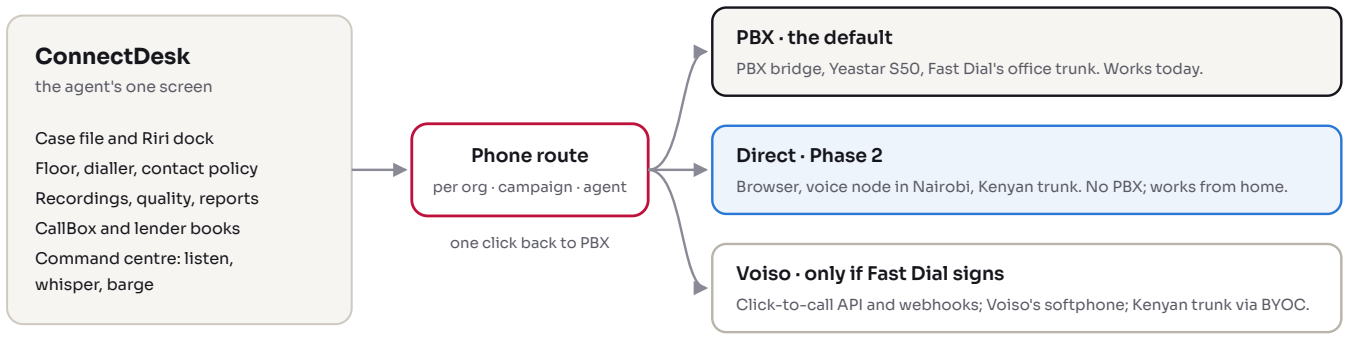
ConnectDesk stays the agent's one screen. What changes is the road a call takes, and where ConnectDesk itself runs.

7 The Phase 2 architecture



Two roads for the same call. Today the S50 in the office carries every call, and ConnectDesk can only steer it from Dublin. In Phase 2 the agent's browser sends audio straight to the voice node in Nairobi, which dials through a Kenyan trunk. ConnectDesk sits in the same data centre as the voice node and the CallBox database. Browser audio is encrypted and travels over port 443, through a TURN relay when a home router blocks it. The trunk carries ordinary telephone audio (G.711 A-law).

The phone route: one setting, three roads



Nothing is all-or-nothing. Each organisation has a default route, and campaigns and agents can override it. A Direct seat that cannot reach the voice node says so and offers the PBX. Supervisors can move the whole floor back to the PBX in one click.

The parts and why we chose them

PART	ITS JOB	TECHNOLOGY	WHY THIS ONE
Agent phone	The agent's extension in the browser: call, hold, mute, keypad, transfer, three-way calls	JsSIP over secure WebSocket; Opus audio; browser echo and noise processing, plus AI noise removal	Already built in ConnectDesk (the Riri dock)

PART	ITS JOB	TECHNOLOGY	WHY THIS ONE
Voice node	Browser gateway, routing, recording, listen/whisper/barge, Riri's line, dialler pacing	**Asterisk 22 LTS**, fully supported to October 2028 and patched to October 2029	The Phase 1 edge is already Asterisk. Ubuntu's Asterisk 20 drops to security fixes only on 19 October 2026. Asterisk 24 LTS is due on 15 October 2026; we move to it once it settles.
TURN relay	Gets audio through home routers and 4G networks that block it	coturn, on the voice node	Free and standard
Kenyan SIP trunks	The road to Safaricom, Airtel and Telkom; caller ID; inbound numbers	A primary carrier plus a second for backup, both delivered to the data centre	A legal local caller ID at local prices
ConnectDesk app	Screens, floor, dialler logic, contact policy, reports	Next.js, as today, moved to Nairobi (section 10)	A few milliseconds from CallBox and the voice node
Control link	ConnectDesk to the voice node: place calls, monitor, receive events	Asterisk REST Interface (ARI) plus signed webhooks	Events arrive as they happen, which ends 1.5-second status polling
Recording store	Recordings and transcripts	Object storage in Kenya. Stereo files keep the agent and the customer on separate channels.	Quality reviews and AI review
Monitoring	Is it up, and how good is the audio?	Health checks, per-call quality statistics, alerts	The only way to tell noise from network faults

In code, Direct is a third *driver* behind ConnectDesk's existing phone-system interface (`src/lib/desk/telephony/types.ts`), beside `yeastar-s` and `simulated`. It implements the same verbs: originate, hang up, hold, transfer, monitor, extensions, live calls, probe and recording. The voice node keeps pulling its seats from ConnectDesk every minute, as the Phase 1 edge does (`bridge/edge`, which becomes `bridge/voice`). Calls on Direct are also written to CallBox's call tables, so CallBox's own reports keep counting them while both systems run.

How each call works

1. **Click-to-call.** The agent presses Call in the case file, and the browser phone dials the voice node over WebSocket. The node checks the seat and the destination (Kenyan numbers only), sets the organisation's caller ID, starts recording and dials the trunk. Ringing, answered and ended reach ConnectDesk as events, so the call record updates instantly. The recording is uploaded at hang-up, and the outcome pad opens.
2. **Auto-dialler (progressive).** ConnectDesk claims the next record for a Ready agent, honouring the contact policy as it does today. The node dials the customer and joins the agent's browser on answer, with a short tone and the customer's name. Busy, no-answer and network announcements are logged and the dialler moves on. *Predictive* pacing (more lines than agents) comes after we have measured connect rates. It will hold abandoned calls under 3% of live calls, the UK regulator's benchmark.
3. **Inbound.** A customer calls Fast Dial's number. The trunk delivers the call to the node, Riri answers with her keypad menu, and the call hunts Ready agents, account owner first. The agent's browser rings with the customer's file. This is the logic of today's 716 line, without the laptop.
4. **Supervisor.** The supervisor picks Listen, Whisper or Barge in the command centre's seat drawer. The node rings the supervisor's browser phone straight into the call in that mode, and the floor log records it. Taking over a call (Voiso's *Intercept*) follows in weeks 5–6.
5. **Transfer and three-way.** These work as in the Phase 1 browser phone (SIP REFER and a local three-way mix).

6. **Failure.** If the node or the trunk is down, the phone says so and offers "Use PBX". Automatic failover follows in weeks 5–6.

AGENTS AT HOME

- **Kit:** a laptop with Chrome or Edge, a USB noise-cancelling headset, and a stable connection of 2 Mbps each way (fibre, with 4G as backup) in a quiet room.
- **No VPN.** Call control runs over HTTPS on port 443. Audio goes direct over UDP, or through the TURN relay over TLS when UDP is blocked.
- **A 20-second check before "Ready":** microphone level, background noise, and the delay, jitter and loss to the voice node. If it fails, the agent cannot go Ready, and the supervisor sees why.
- **A second sign-in factor** for remote agents. Fast Dial signs in with a password only today because there is no SMS provider; an authenticator app fixes that. Agents can never download recordings, and idle screens lock.

SECURITY AND FRAUD

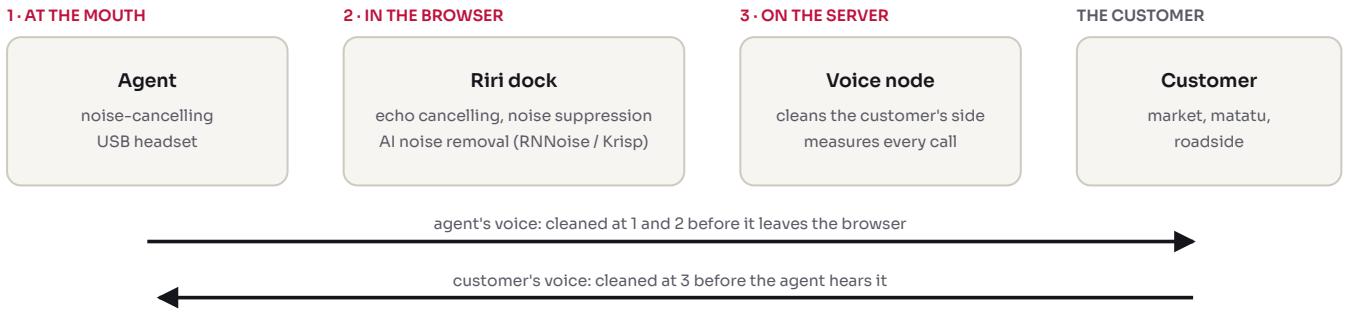
- No SIP port is open to the internet except to the carriers' addresses. Browsers reach the node only over secure WebSocket, using per-seat credentials that ConnectDesk generates and rotates.
- Calls go only to Kenyan mobile and landline numbers, and international dialling is off. Each trunk has daily caps on minutes and spend, with alerts.
- Browser audio is encrypted (DTLS-SRTP), as is trunk audio where the carrier offers it. Recordings are encrypted at rest, and every play and download is audited, as today.
- Recordings and call data stay in Kenya, so lenders' borrower data never needs to leave the country.

8 The noise problem: calls that sound like a market

A colleague's earlier FreeSWITCH build connected calls, but the room came through louder than the people. It is the most common complaint about software phones in call centres. It is not FreeSWITCH's fault, and it is not FreeSWITCH's job: **a telephone switch carries audio, it does not clean it.** Asterisk is no different. The cleaning has to be designed in, and it happens in three places.

Why it happens, in order of how often we expect it

CAUSE	WHAT IT SOUNDS LIKE, AND THE FIX
1 · The microphone hears the room	Laptop microphones and cheap headsets pick up everything. On an open floor the loudest noise is other agents' voices. **Fix:** USB headsets with a noise-cancelling boom microphone. No laptop microphones, no speakers.
2 · Speakers and no echo control	Sound from a speaker goes back into the microphone, so the customer hears themselves and the room. **Fix:** headsets only, and the browser's echo cancellation switched on.
3 · Volume turned up	Boosting a weak microphone boosts the room too, and automatic gain "pumps" the noise up in every pause. The open-source CRM we reviewed doubles the volume of every call (`VOLUME(RX)=2`). **Fix:** fix the microphone, not the volume. Set gain once.
4 · The customer is in a market	Many borrowers answer from matatus, markets and roadsides. **Fix:** clean the customer's side on the server before the agent hears it.
5 · It is the network	Packet loss and jitter make audio choppy or robotic, and people report it as noise. **Fix:** measure every call (jitter, loss, delay, MOS) so the cause is known, not guessed.



Clean audio at the source, in the browser and on the server. The switch in the middle only carries the audio, which is why replacing one switch with another never fixed the noise.

LAYER	WHAT WE DO	COST
1 · At the mouth	Noise-cancelling USB headsets with a dual-microphone boom on every seat, and a headset standard for home agents. The pre-shift check measures the background noise and blocks "Ready" when it is too loud.	About KES 6,000–15,000 per headset, once
2 · In the browser	The browser's echo cancellation, noise suppression and gain control, switched on. Then AI noise removal runs in an audio worklet before the voice leaves the PC. **RNNoise** (free; it handles steady noise such as fans and traffic, and Jitsi ships it the same way) or **DeepFilterNet** (free; better quality but heavier). For other people's voices on the floor, the real "market" noise, use **Krisp**'s background-voice cancellation. In one published test it cut transcription errors on noisy audio from 117.6% to 23.5%, and it adds about 15–25 ms of delay.	RNNoise and DeepFilterNet free. Krisp's SDK from $5,000 a year, or its call-centre app from $10 per agent a month, which works with any softphone, MicroSIP included.
3 · On the server	Asterisk's `DENOISE()` on the customer's side of the call (mild, built in). Then a stronger AI clean-up of the customer's audio, tried on recordings before it goes live.	Free; server CPU only
Measure it	Each call stores the browser's own quality statistics (jitter, loss, round-trip time, estimated MOS). Recordings get an objective noise score, and Riri's after-call review flags noisy calls for the supervisor.	Built into Phase 2

THE "MARKET TEST"

Record 20 calls per group on the busiest floor hour, before and after each change. Score them with an objective noise model and with Riri's transcription error rate, and have agents rate each call from 1 to 5. A change goes to the whole floor only if the noise score improves and nobody hears extra delay.

QUICK WIN THIS WEEK, EVEN ON THE PBX

Give the five noisiest seats proper headsets now, and trial the Krisp call-centre app on them. It works as a virtual microphone with MicroSIP, so the PBX route benefits before Phase 2 ships. Ask Voiso to demonstrate the same: noise removal on the agent's side, on the customer's side, and of background voices.

9 Lessons from earlier attempts

From FreeSWITCH-from-scratch builds, including a colleague's

We have not seen that code. These are the usual failure points, and how Phase 2 avoids each one.

- **The switch gets built before the job.** Weeks go into SIP plumbing while the agent screen, supervisor tools and dialler wait. *Ours:* the job already exists in ConnectDesk; the engine only replaces the PBX's transport.
- **Calls connect but sound wrong.** Behind NAT, a wrong external address gives one-way audio. A bad WebSocket certificate breaks the whole profile, and home users without a TURN relay get silence. *Ours:* fixed public addresses in a data centre, real certificates and a TURN relay from day one.
- **Noise left to the switch.** See section 8.
- **Packaging.** FreeSWITCH's official packages now need a SignalWire access token, and source builds are slow and drift. *Ours:* Asterisk 22 LTS with a pinned build kept in the repo.
- **No measurement and no fallback.** *Ours:* per-call quality statistics, and the PBX one click away.

From the open-source CRM we reviewed (`callcenter-crm`)

It uses Asterisk and a Zadarma SIP trunk.

- **Worth keeping:** SIP over TLS on port 8443 to get past ISPs that block 5060; Fail2ban against password guessing; an address allow-list on the trunk; a hang-up webhook to the CRM; Opus on the operator's leg; an adaptive jitter buffer.
- **Gaps for us:** operators still use MicroSIP, and its roadmap lists the browser phone, queues and a live dashboard as not done. It doubles every call's volume, which amplifies the room. Its trunk is foreign, and in Kenya a foreign trunk showing a +254 number is blocked.

From our own Phase 1

- Use one fixed address per PBX session. Vercel's rotating addresses multiplied PBX events 10–25 times until the bridge on .228 fixed it.
- An engine on a laptop goes down when the laptop sleeps. The voice node lives in a data centre.
- Polling a small PBX makes it stall, so Phase 2 is event-driven.

10 Hosting: the cheapest setup that runs Phase 2 well

What runs where today

PIECE	WHERE IT RUNS	NOTE
ConnectDesk, Lending Console, Micro Eazy APIs	One Next.js app on Vercel (Pro plan), region Dublin	Every desk screen is served from Ireland
ConnectDesk's own tables	Supabase Postgres, Ireland	Users, seats, calls, floor and settings
Fast Dial's CallBox data	Postgres on server .228 (Servercore, Nairobi)	Beside CallBox, the PBX bridge and the SQL relay
Micromart's books	Micromart's SQL Server, reached through the relay	In Kenya

"The database is on .228" is true for CallBox, but not for ConnectDesk's own tables. Every desk screen is built in Dublin from data in Nairobi *and* Ireland. That is slow, and on Vercel every request is billed.

What Vercel is costing, measured

We read Vercel's usage report at 05:47 on 8 October. Each Vercel day runs from 10:00 to 10:00 Nairobi time. Of the usage, 99% comes from the app above (Vercel project `birgen-ai-lms`).

VERCEL DAY	USAGE COST	FUNCTION CALLS	CPU HOURS	MEMORY GB-H	DATA OUT	CDN REQUESTS
Sun 4 Oct	$1.61	134,000	1.3	45	1.0 GB	78,000
Mon 5 Oct (PBX down 5 h)	$5.39	363,000	7.5	106	3.2 GB	183,000
Tue 6 Oct	$9.20	765,000	12.5	123	5.8 GB	388,000
Wed 7 Oct (to 05:47 Thu)	$4.24	382,000	4.4	53	3.8 GB	193,000

What drives it. Each agent's screen holds a live floor stream that reconnects every 5 minutes. During a call it asks for the call's status every 1.5 seconds. Every request costs an invocation, CPU, memory and an observability event. At today's 10–15 people that comes to $5–9 of usage per working day, and Vercel's $20 monthly usage credit lasts about three days.

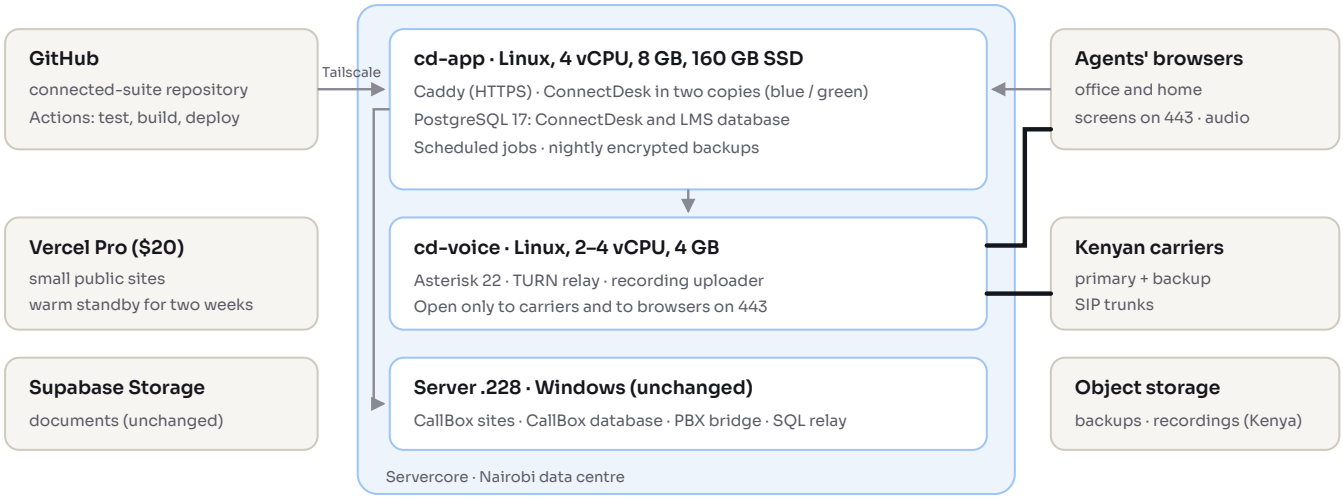
At 30 agents (20 Fast Dial and 10 Micromart), expect 2–3 times today's traffic: about $11–28 of usage per working day, or **$300–700 a month**, plus the $20 Pro seat. Cutting the polling, which Phase 2's event-driven design does anyway, would bring it to about $150–250. There is also a CDN tier on top. Pro includes 1 million CDN requests a month under Vercel's flat-rate CDN, and a month over capacity moves the team to the $20 tier (10 million) or the $100 tier (50 million) from the next billing cycle.

> CAN A SEPARATE FREE VERCEL PROJECT CARRY THE PILOT FOR A MONTH?
>
> **No, for two reasons.** First, Vercel's free Hobby plan is for *personal, non-commercial* use, and a pilot for paying call centres is commercial. Second, plans and limits belong to the account, not the project, so a new project in the same account changes nothing. A new free account would hit Hobby's hard caps within days: **4 CPU hours a month** (ConnectDesk used 12.5 in one day), **1 million function calls** (765,000 in one day) and **1 million CDN requests** (388,000 in one day). When a cap is hit, the project stops serving until the next month, so the floor would go dark mid-pilot.

The options

OPTION	A MONTH	TRADE-OFF
1 · Vercel Pro, as is	$300–700	No work, but served from Dublin, on rotating addresses, and the cost grows with every agent.
2 · Vercel Pro, polling cut	$150–250	About a week of code. Still served from Dublin.
3 · On server .228 itself	KES 0 extra	Cheapest, but .228 is a Windows server already running six CallBox sites and CallBox's database for several call centres. A ConnectDesk spike could slow theirs, and Asterisk cannot run on Windows anyway. Use only as a short stop-gap.
4 · Two Linux servers in the same Nairobi data centre	KES 16–20k about $125–155	**Recommended.** Flat cost up to roughly 60–80 agents. A few milliseconds from CallBox. Fixed addresses. The voice node needs a Linux server anyway.
5 · Option 4, plus ConnectDesk's database moved from Ireland to Nairobi	no extra	**Recommended, on the same weekend.** Every query becomes local. Supabase file storage (documents) stays where it is; only the Postgres database moves.

Keep Vercel Pro ($20 a month, covered by its usage credit) for the small public sites, and as a warm standby for two weeks after the move.



The Phase 2 hosting layout. Everything that talks to everything else sits in one Nairobi data centre. GitHub deploys over the private Tailscale network, so no server exposes SSH to the internet. Thick lines carry call audio.

Continuous deployment: push to main, live in about five minutes, one command to roll back

1. **Two small code changes.** Add `output: "standalone"` to `next.config.ts` and a `/api/health` route that answers 200 when the app can reach its database. Neither exists today. Nothing else in the app depends on Vercel: there are no `@vercel/*` packages, and the crons already authenticate with `CRON_SECRET`.
2. **The workflow** (below). The build job installs dependencies, type-checks, runs the fast test suites, builds and packages. The deploy job joins the tailnet with a short-lived key, copies the release and runs `cd-deploy` on the server.
3. **Blue/green on one server.** `cd-deploy` starts the new release on the idle port (3001 or 3002) and waits for `/api/health`. Then it points Caddy at the new port and stops the old copy 30 seconds later; live streams reconnect by themselves. It keeps the last five releases, and `cd-deploy --rollback` switches back.
4. **Database changes never run automatically.** A separate, manual "migrate" workflow takes a backup first, then applies the reviewed SQL.
5. **Secrets** live only on the server (`/etc/connectdesk/env`, readable by the service account). GitHub holds only the deploy credentials. Build-time `NEXT_PUBLIC_*` values go in GitHub's repository variables.
6. **Scheduled jobs.** Vercel's seven daily crons become systemd timers that call the same `/api/cron/*` URLs at the same UTC times, with `CRON_SECRET`. The PBX bridge keeps calling `/api/cron/desk-floor` every minute.
7. **Watching it.** An external uptime check on `/api/health` every minute, with SMS and e-mail alerts. Logs go to journald, with alerts on disk and CPU.
8. **Backups.** A nightly encrypted `pg_dump` plus continuous WAL archiving to object storage, so any minute can be restored. Weekly server snapshots. A restore test every month.
9. **The voice node is code too.** Its pinned Asterisk build and base configuration live in the repo (`bridge/voice`) and deploy through the same pipeline. Seats and routes keep coming from ConnectDesk every minute.

```yaml
# .github/workflows/deploy-nairobi.yml   (connected-suite)
name: deploy-nairobi
on: { push: { branches: [main] }, workflow_dispatch: {} }
concurrency: { group: production, cancel-in-progress: false }
jobs:
  build:
    runs-on: ubuntu-24.04
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 24, cache: npm }
      - run: npm ci
      - run: npx tsc --noEmit
      - run: npm run test:dialler && npm run test:edge && npm run test:call-placement
      - run: npm run build                # prisma generate && next build (standalone)
      - run: |
          mkdir out && cp -r .next/standalone/. out/
          cp -r .next/static out/.next/static && cp -r public out/public
          tar -C out -czf release.tgz .
      - uses: actions/upload-artifact@v4
        with: { name: release, path: release.tgz, retention-days: 14 }
  deploy:
    needs: build
    runs-on: ubuntu-24.04
    environment: production             # optional: require a reviewer
    steps:
      - uses: actions/download-artifact@v4
        with: { name: release }
      - uses: tailscale/github-action@v3   # joins the tailnet; no public SSH
        with:
          oauth-client-id: ${{ secrets.TS_OAUTH_CLIENT_ID }}
          oauth-secret: ${{ secrets.TS_OAUTH_SECRET }}
          tags: tag:ci
      - run: |
          scp release.tgz deploy@cd-app:/srv/connectdesk/incoming/$GITHUB_SHA.tgz
          ssh deploy@cd-app sudo /usr/local/bin/cd-deploy $GITHUB_SHA
```

```bash
#!/usr/bin/env bash   # /usr/local/bin/cd-deploy on cd-app (outline)
set -euo pipefail
SHA=$1; ROOT=/srv/connectdesk; REL=$ROOT/releases/$SHA
mkdir -p "$REL" && tar -xzf "$ROOT/incoming/$SHA.tgz" -C "$REL"
LIVE=$(cat $ROOT/live 2>/dev/null || echo green)            # blue = :3001, green = :3002
NEXT=$([ "$LIVE" = blue ] && echo green || echo blue); PORT=$([ "$NEXT" = blue ] && echo 3001 || echo 3002)
ln -sfn "$REL" "$ROOT/$NEXT" && systemctl restart "connectdesk@$NEXT"
for i in $(seq 1 30); do curl -fsS "http://127.0.0.1:$PORT/api/health" >/dev/null && break; sleep 2; done
curl -fsS "http://127.0.0.1:$PORT/api/health" >/dev/null || { systemctl stop "connectdesk@$NEXT"; exit 1; }
echo "reverse_proxy 127.0.0.1:$PORT" > /etc/caddy/upstream.caddy && systemctl reload caddy
echo "$NEXT" > $ROOT/live; sleep 30; systemctl stop "connectdesk@$LIVE"
ls -1dt $ROOT/releases/* | tail -n +6 | xargs -r rm -rf           # keep five releases
```

Each copy runs as `connectdesk@blue` or `connectdesk@green`, a systemd template that runs `node server.js` as an unprivileged user, from `/srv/connectdesk/%i`, with `EnvironmentFile=/etc/connectdesk/env` plus a per-colour file holding `PORT`. Caddy's site block for `connectdesk.`, `lms.` and the API hosts imports `/etc/caddy/upstream.caddy`.

> **THE MOVE, ONE WEEKEND**
>
> **Week 1:** build both servers and the pipeline, restore a copy of the database, and run the full app against it on a staging address. Compare page times with Vercel and rehearse the cut-over twice. **Saturday 24 October, 21:00–23:00:** lower the DNS TTL two days ahead. At 21:00 announce a read-only window, take the final dump from Supabase and restore it in Nairobi. Then switch `DATABASE_URL`, switch DNS and run the smoke tests. Vercel stays deployed, pointing at the Nairobi database, as a warm standby. **Rollback:** point DNS back to Vercel. If writes had already happened in Nairobi, copy them back with a reverse dump.

11 The plan, week by week

TRACK	W0 · 8–11 OCT	W1 · 12–18 OCT	W2 · 19–25 OCT	W3 · 26 OCT–1 NOV	W4 · 2–6 NOV	W5–6 · 9–20 NOV
Fast Dial	Voiso told: monthly only. Trunk quotes asked. Pilot agents named.	Voiso technical demo. Trunk ordered.	Trunk delivered (target)	—	**Decision, Fri 6 Nov**	Sign or stop Voiso. Scale plan.
Voiso test	Questions sent	Demo; trial set up on Fast Dial's numbers	Test calls; costs measured	Test calls; supervisor and integration tests	Scorecard	—
Hosting	Servers ordered	Servers, pipeline, staging copy, two rehearsals	Cut-over Sat 24 Oct	Vercel as standby	Vercel retired for the app	—
Voice engine	—	Asterisk 22, browser seats, Direct driver, phone-route setting	First calls on a test trunk; recording; call records	3 office + 2 home agents on the real trunk; supervisor tools	10–20 agents; inbound line on the node	Predictive pacing, AMD, automatic fallback, second node
Audio quality	5 headsets	Browser processing; pre-shift check	AI noise removal in the browser	Customer-side clean-up; Krisp trial on 5 seats	Market test, before and after	Roll out what worked

If the carrier is slow, first calls run on a test trunk from Africa's Talking. It opens quickly by allow-listing the server's address, and it needs a small callback endpoint, which ConnectDesk provides. Micromart stays on its PBX by default. Direct for Micromart needs Micromart's own trunk and caller ID, and is offered after Fast Dial's pilot passes.

12 Risks and how we contain them

RISK	LIKELIHOOD	WHAT WE DO
Carrier slow to deliver a trunk to the data centre	Medium	Ask two carriers now. Meanwhile test on Africa's Talking's SIP, and keep the PBX as the default.
Audio at some homes worse than the PBX	Medium	The pre-shift check and the TURN relay. Agents who fail work from the office or on the PBX.
Noise not solved by software alone	Medium	Headsets first, measure every change, trial Krisp.
Voice node outage	Low	One click back to the PBX, monitoring and alerts, and a second node in weeks 5–6.
Self-hosting adds operations work	Medium	Automated deploys, backups and alerts, written runbooks, and Vercel as a warm standby for two weeks.
Data loss during the database move	Low	Two rehearsals, a backup first, a short read-only window, and a reverse plan.
Stolen seat credentials used for toll fraud	Low	Kenyan numbers only, spend caps, alerts, and credentials that rotate.
Fast Dial signs Voiso anyway	—	A ConnectDesk adapter: click-to-call API, webhooks for outcomes, and recordings copied to storage Fast Dial controls. ConnectDesk stays the agent's workspace.

A Questions for the Voiso demo

Ask for the answers in writing. Prices and routes said aloud in a demo are not commitments.

#	QUESTION	A GOOD ANSWER	A RED FLAG
Kenya, caller ID and minutes			
1	When an agent calls a Safaricom or Airtel number, which carrier carries the call, and is the call started inside Kenya?	A named Kenyan-licensed partner, or Fast Dial's own trunk	"International route"; "caller ID not guaranteed"
2	Which number will the customer see? Show it on a Safaricom and an Airtel handset during the demo.	Fast Dial's own Kenyan number, every time	A foreign or random number; calls that never ring
3	Can we plug in our own Kenyan SIP trunk (BYOC)? Where does it connect (which data centre and address)? Is there a fee? Who supports it?	Yes, with a named data centre and engineer	Only Voiso's carriers allowed
4	What is your per-minute rate to Safaricom, Airtel and Telkom mobiles and to Kenyan landlines? Per second or per minute? What are the AMD and number-checking charges?	A written rate card in KES or USD	"Destination-specific", with no card
5	Can Fast Dial's existing inbound number ring Voiso agents, and can we keep the number?	Yes, through our trunk	Only by porting to Voiso
Audio and location			
6	Which data centre would carry our calls, and what is the measured round trip from Nairobi? On BYOC calls, does the audio go Kenya, then your data centre, then back to Kenya?	Under 150 ms one way, measured	No measurement offered
7	Do you remove noise on the agent's side, on the customer's side, and background voices? Show it on a noisy floor.	Yes, live, all three	"Use a good headset"
8	Which codecs do you use for browser agents and for trunks? How do you handle packet loss?	Opus with error correction; G.711 to trunks	No clear answer
9	Can we get per-call quality figures (MOS, jitter, loss) in the CDR or the API?	Yes	No
Supervisors and agents			
10	Show Monitor, Whisper, Barge-in and Intercept on a live call. Can ConnectDesk trigger them through the API?	Yes, with documented endpoints	Only inside Voiso's dashboard
11	Can the softphone live inside ConnectDesk (iframe or SDK) so agents use one screen? Which browsers work?	Yes, demonstrated in our app	Agents must keep two windows open; Chrome only
12	Can we control hold, transfer, conference, mute and keypad tones through the API?	Yes	Start and stop only
13	Can we read and set agent states (ready, break, wrap-up) through the API and webhooks?	Yes, both ways	Read-only
Dialler			
14	How does predictive pacing work? What abandoned-call cap do you apply, and how is it measured? How accurate is AMD on Kenyan networks, where operators play recorded messages in English and Swahili?	A configurable cap (3%) with reports; Kenyan tuning	No cap; AMD untested in Kenya

#	QUESTION	A GOOD ANSWER	A RED FLAG
15	Can ConnectDesk push contacts and receive each attempt's outcome? Can we enforce our contact policy (caps per customer, quiet hours)?	Yes, through the API and webhooks	CSV uploads only
Data, contract and support			
16	In which country are recordings and call records stored? Can recordings be kept only in our own storage? What are the retention and deletion controls?	Our storage only, deletion on request	Copies kept abroad regardless
17	Which certifications do you hold (ISO 27001, SOC 2)? Will you sign a data processing agreement under Kenya's Data Protection Act? Can we see your list of sub-processors?	Yes, all three	A GDPR statement only
18	When we leave, how do we export call records, recordings and contacts, in what format, and at what cost?	Free and complete	Paid or partial
19	What do 22 Pro users cost monthly versus annually? Any minimums? Can the trial run 30 days on our numbers?	Monthly; trial extended	Annual contract required to test
20	What support hours do you offer in Nairobi time? Do you have a Kenyan partner? What is the uptime SLA and the credit if it is missed? What were the outages of the last 12 months?	A written SLA and a status page	No SLA
21	Which Kenyan or East African customers can we call as references?	Two or more	None

Tests to run during the trial

1. 100 calls to Safaricom and Airtel numbers: the connect rate, the caller ID seen, and the cost per minute on the invoice.
2. Round-trip time and MOS from Fast Dial's office and from one home connection.
3. Monitor, Whisper and Barge-in on 10 live calls.
4. Click-to-call from a ConnectDesk case file through the API, with the outcome and recording returned to ConnectDesk.
5. The market test (section 8) on Voiso's audio.

B Acceptance tests for Direct calling

#	TEST	PASS MARK
1	20 calls each to Safaricom and Airtel test handsets	Fast Dial's number shown on every call
2	200 consecutive dial attempts	No more than 1% technical failures (busy and no-answer excluded)
3	Two-way audio on every answered call in the pilot week	No case of one-way audio
4	Audio quality	Average MOS 4.0 or better in the office and 3.8 or better at home; one-way delay under 150 ms; packet loss under 1%
5	Listen, whisper and barge from the command centre on a live call	Connected within 3 seconds. In whisper the agent hears the supervisor and the customer does not.
6	Recordings	In ConnectDesk within 60 seconds of hang-up, with agent and customer on separate channels

#	TEST	PASS MARK
7	Inbound: call Fast Dial's number	Riri answers. Pressing 1 rings a Ready agent's browser with the customer's file open.
8	Remote agents	One agent on home fibre and one on 4G complete a full shift
9	Load	20 calls at once with the voice node under 50% CPU
10	Fallback	With the voice node stopped, the floor is back on the PBX within 2 minutes
11	Security	A port scan from the internet finds no open SIP. A call to a non-Kenyan number is refused and raises an alert.
12	Noise	The market test shows a better noise score, and nobody reports extra delay

C Cost workings and assumptions

ITEM	WORKING
Exchange rate	KES 129 to US$1 (check on the day)
Minutes	In the 30 days to late September, Fast Dial answered 30,525 outbound calls (about 1,000 a day). In the week to 6 October the floor talked for about 94 hours, which is about 24,000 talk minutes a month. Rounding each call up to the minute pushes billed minutes to 25,000–40,000, depending on the route. We use 30,000 for every option.
International route	Published rates to Kenyan mobiles: Twilio $0.3933 a minute; Plivo $0.448 a minute (Safaricom and Airtel, billed by the minute). 30,000 × $0.39–0.45 = $11,700–13,500, or KES 1.51–1.74M.
Kenyan trunk	KES 2.50–4.00 a minute from published Kenyan voice and SIP rates, for example Africa's Talking's outgoing rate. 30,000 × KES 2.50–4.00 = KES 75,000–120,000. Replace with Fast Dial's actual Safaricom rate and the new quotes.
Voiso licences	Pro at $65 per user per month (monthly) or $52 (annual) × 22 users (20 agents and 2 supervisors) = $1,430 (KES 184k) or $1,144 (KES 148k). AMD, caller-ID and number fees are extra and not included.
Servers	Published Kenyan cloud prices put an 8 GB server at about KES 11,500 and a 4 GB server at about KES 4,000–5,000 a month. With object storage and backups, the total is KES 16,000–20,000. Confirm Servercore's Nairobi prices before ordering.
Optional AI noise removal	The Krisp call-centre app from $10 per agent a month on annual billing: 20 × $10 = $200 (KES 26k).
Vercel	Measured usage of $1.61–9.20 per Vercel day (4–7 October) at 10–15 users, scaled ×2–3 for 30 agents over 26 working days: $280–720, rounded to $300–700. Pro is $20 per developer seat per month, including $20 of usage credit. CDN tiers are $20 for 10M requests and $100 for 50M.

D Sources

Retrieved 8 October 2026 unless stated. Internal figures come from ConnectDesk's Phase 1 records (the CallBox call tables, ConnectDesk call records and the PBX bridge's health log) and from Vercel's usage report read at 05:47 EAT on 8 October.

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